

LAB 04

Title: Checking Left Recursion in a Context-Free Grammar Using C++ with Debugging Techniques

Objective

To learn how to debug a C++ program using Breakpoints and the Watch Window, and to determine whether a given grammar production contains left recursion.

Procedure

1. Complete environment setup.
2. Write the left recursion checking code.
3. Set break points.

Breakpoint Location	Reason
<code>cin >> p;</code>	To verify that the grammar production is entered correctly.
<code>int pos = p.find("->");</code>	To verify that the -> operator is found and the right-hand side starts from the correct position.
<code>for (int i = pos + 2; i < p.length(); i++)</code>	To observe the traversal of each symbol in the right-hand side (RHS) of the production.
<code>if (p[i] == left)</code>	To check whether the current symbol matches the left-hand side non-terminal.
<code>found = true;</code>	To confirm that a left recursive production has been detected.

4. Execute with Next Line.
5. Track the changes on Watch.

Code with break-points

```

LeftRecursionValidation.cpp
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      string p;
7
8      cout << "Enter a production: ";
9      cin >> p;
10
11     char left = p[0];
12     int pos = p.find("->");
13
14     bool found = false;
15

```

```

16   for (int i = pos + 2; i < p.length(); i++)
17   {
18       if (p[i] == left)
19       {
20           if (i == pos + 2 || p[i - 1] == '|')
21           {
22               found = true;
23               break;
24           }
25       }
26   }
27
28   if (found)
29       cout << "Left Recursive";
30   else
31       cout << "Not Left Recursive";
32
33   return 0;
34 }

```

Output

